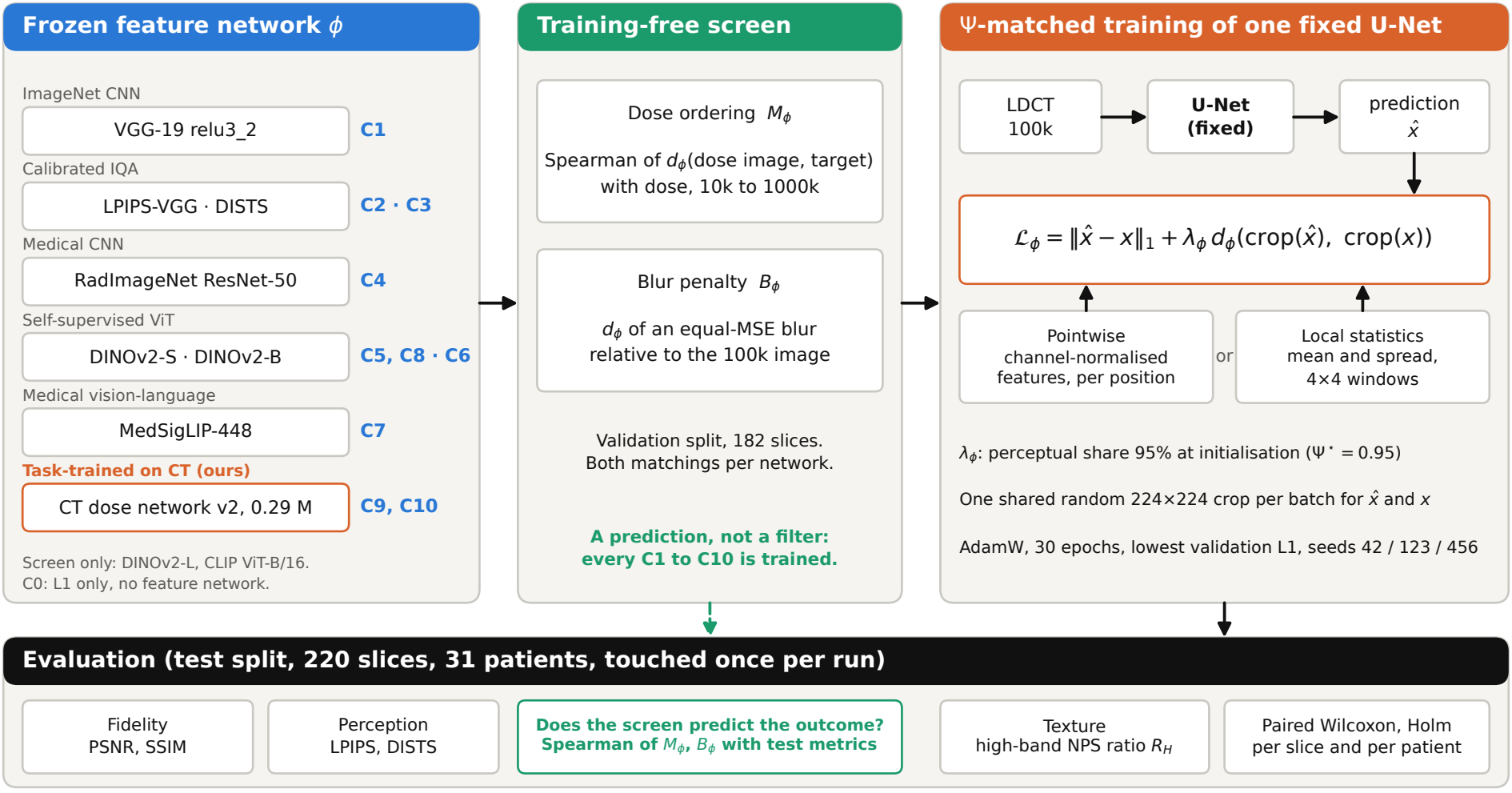
Frozen feature network ϕ
ImageNet CNN
VGG-19 relu3_2
C1
Calibrated IQA
LPIPS-VGG · DISTS
C2 · C3
Medical CNN
RadImageNet ResNet-50
C4
Self-supervised ViT
DINOv2-S · DINOv2-B
C5, C8 · C6
Medical vision-language
MedSigLIP-448
C7
Task-trained on CT (ours)
CT dose network v2, 0.29 M
C9, C10
Screen only: DINOv2-L, CLIP ViT-B/16.
C0: L1 only, no feature network.
Training-free screen
Dose ordering M_ϕ
Spearman of d_ϕ(dose image, target) with dose, 10k to 1000k
Blur penalty B_ϕ
d_ϕ of an equal-MSE blur relative to the 100k image
Validation split, 182 slices.
Both matchings per network.
A prediction, not a filter:
every C1 to C10 is trained.
Ψ-matched training of one fixed U-Net
LDCT 100k
U-Net (fixed)
prediction $\hat{x}$
$\mathcal{L}_\phi = \|\hat{x} - x\|_1 + \lambda_\phi\, d_\phi(\mathrm{crop}(\hat{x}),\ \mathrm{crop}(x))$
Pointwise channel-normalised features, per position
or
Local statistics mean and spread, 4×4 windows
λ_ϕ: perceptual share 95% at initialisation ($\Psi^\star = 0.95$)
One shared random 224×224 crop per batch for $\hat{x}$ and x
AdamW, 30 epochs, lowest validation L1, seeds 42 / 123 / 456
Evaluation (test split, 220 slices, 31 patients, touched once per run)
Fidelity
PSNR, SSIM
Perception
LPIPS, DISTS
Does the screen predict the outcome?
Spearman of M_ϕ, B_ϕ with test metrics
Texture
high-band NPS ratio R_H
Paired Wilcoxon, Holm
per slice and per patient